Hormones and Recombinant DNA Technology

Recombinant DNA Technology

- Process of splicing DNA from one organism into another organism
- Variety of uses.
- Endocrine System uses of recombinant DNA Technology:
  - Synthetic Insulin
  - Synthetic Growth Hormone

Term	Definition
Blunt ends	The ends produced by a straight cut of a sequence of nucleotide bases
Ligase	An enzyme capable of combining two small components of single-strand DNA into one single structure
Phage	Or bacteriophage; a virus that infects bacteria
Plasmid	Small circular strand of DNA distinct from the main bacterial genome; it is composed of only a few genes and is able to replicate independently within a cell
Restriction enzyme	Enzyme that cuts a strand of DNA at a specific sequence of nucleotides
Staggered cut	Produced when a restriction enzyme creates fragments of DNA with unpaired nucleotides that overhang at the break in the strands; called sticky ends
Straight cut	Produced when a restriction enzyme makes a clean break across the two strands of DNA so that the ends terminate in a base pair; called blunt ends
Sticky ends	The overhanging ends produced by a staggered cut of a sequence of nucleotide bases; can be called cohesive ends
Vector	A bacterial plasmid, viral phage or other such agent used to transfer genetic material from one cell to another

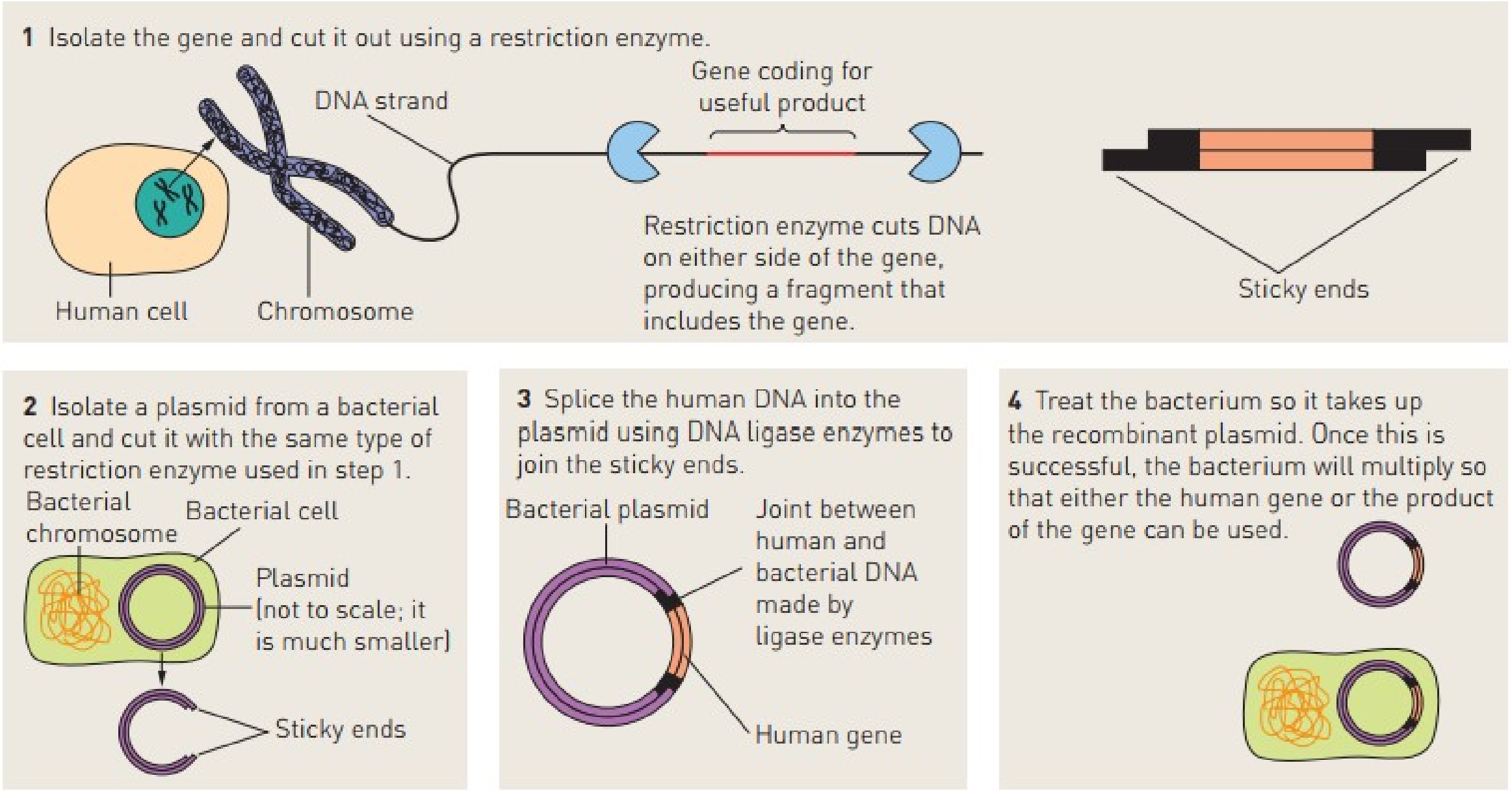

Figure 13.9 A simplified diagrammatic representation of recombinant DNA technology

Examples of use

- Insulin
  - Pre-1982, diabetics treated with insulin from pig/cattle pancreas
  - 1982: insulin produced from genetically engineered bacteria approved.
  - Human gene spliced into bacterial cells, so bacteria would then produce insulin
  - Insulin now produced in huge vats of bacteria.

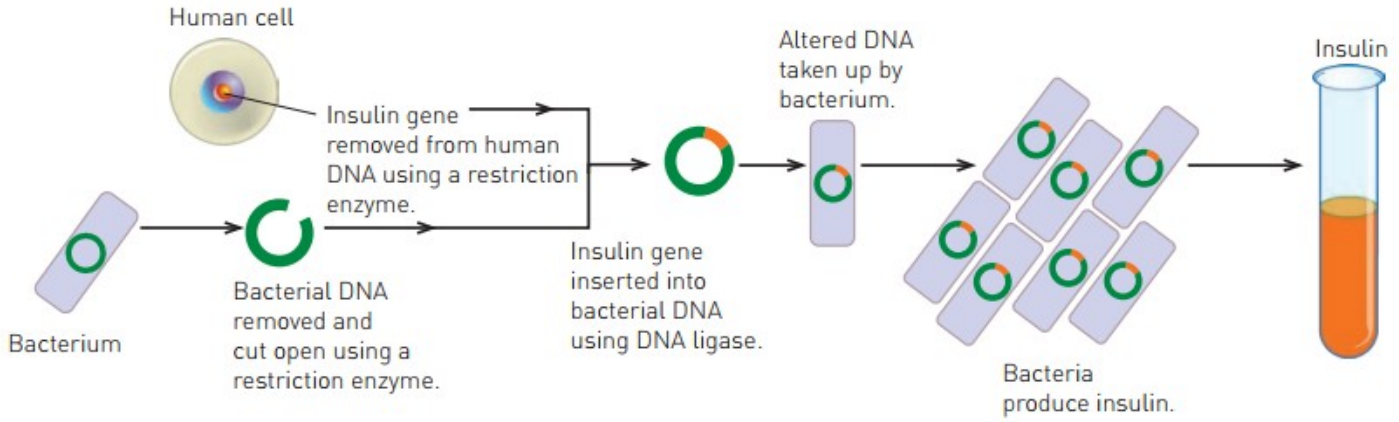

Figure 13.10 A diagrammatic representation of the production of human insulin using recombinant DNA technology

Examples of use cont…

- Human Growth Hormone
  - Now produced by genetically engineered *E. coli* bacteria
  - Used to treat lack of growth hormone in pituitary disorders
  - Also used in cattle to increase milk production